

Name: _____

Date: _____

Class: _____

MyDesign Portfolio

Element D Initial Template AT - 15 points

Element D: Design Concept Generation, Analysis, and Selection

Design Idea Generation (D1)

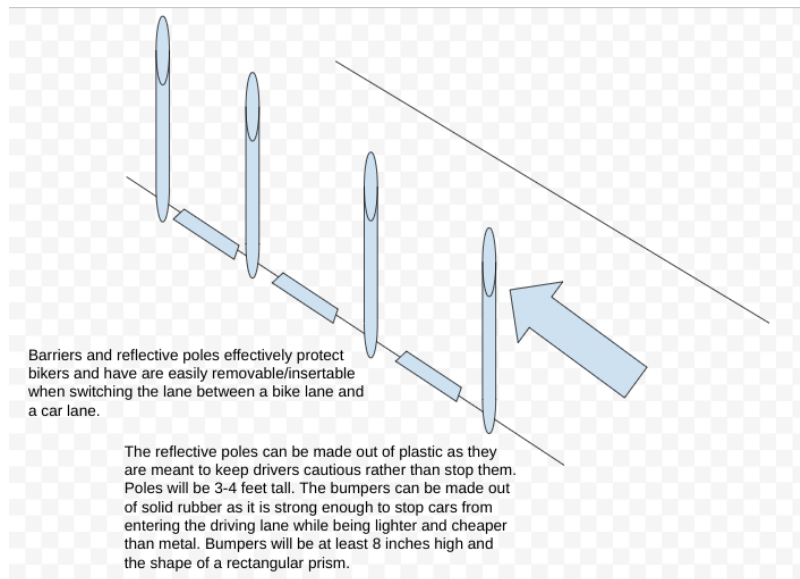
Design Idea #1	Protected Bike Lane
Date:	October 22, 2025
Description:	A bike lane that has raised concrete curbs to separate the main road from the bike lane and small plastic bollards attached on top of the curb to reinforce a divide between the lanes.

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Drawing that includes distinguishing information such as scale, dimensions, and materials:



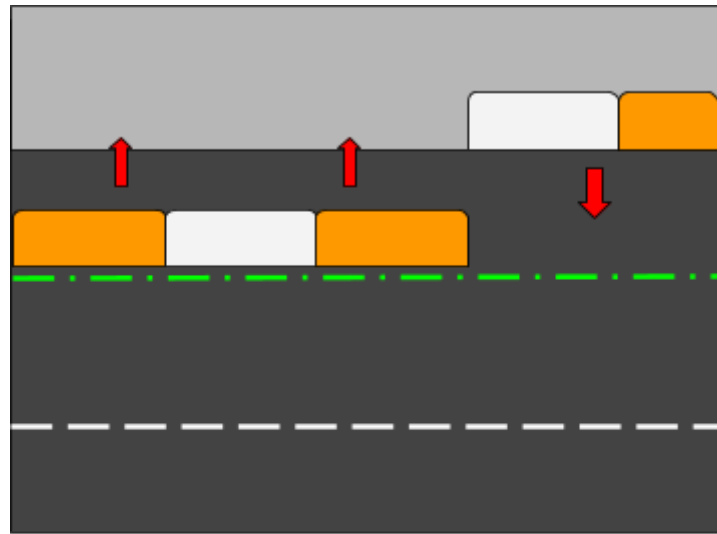
Design Idea #2	Changeable bike lanes
Date:	10/22/25
Description:	Bike lanes on busy roads that can have the barriers moved to make the lane another lane for cars rather than bikes when there is traffic.

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Drawing that includes distinguishing information such as scale, dimensions, and materials:



Barriers with anchors

Design Idea #3	Raised Bike Lanes
Date:	10/22/25

Name: _____

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Description:	The bike lanes on busy roads would be raised above the regular roads. This would make it so that the bikers would be protected and the bike lanes would be super visible because it would be like a curb.
Drawing that includes distinguishing information such as scale, dimensions, and materials:	Side by Side The bike lanes would be around 6 inches tall  Length is the whole road for however the bike lane lasts

Design Idea #4	<i>billards</i>
Date:	<i>Oct 28</i>
Description:	<i>Steel billiards to protect the bikers. Right next to the road so that bikers can feel safe from the cars and are assured not to be hit.</i>

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Drawing that includes distinguishing information such as scale, dimensions, and materials:



Design Idea Comparison and Selection (D1, D2)

(List your requirements for Element C below.)	Weight (multiplied by each design score)	Design 1	Design 2	Design 3	Design 4	Design 5
Safety lines with bollards	10	Yes	Yes	No	Yes	

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Wider bike lanes on the road.	10	Yes	Yes	No	No	
buffers but making them bigger 5 feet wide and 3 feet tall separating	5	Yes	Yes	No	Yes	
The bike lanes must include bright reflective paint and LED lighting placed 30 feet apart	10	Yes	No	No	No	
Install clear signage every 100 feet and at every intersection	5	No	No	No	No	
Total score		35/40	25/40	0/40	15/40	



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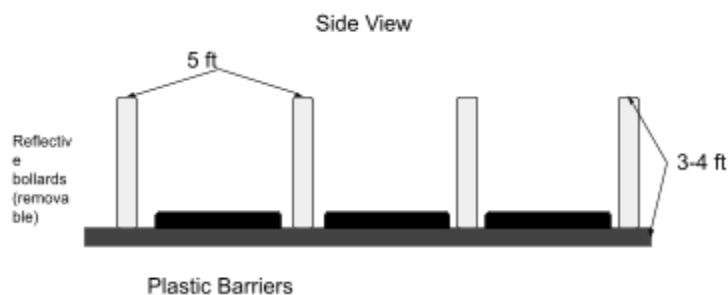
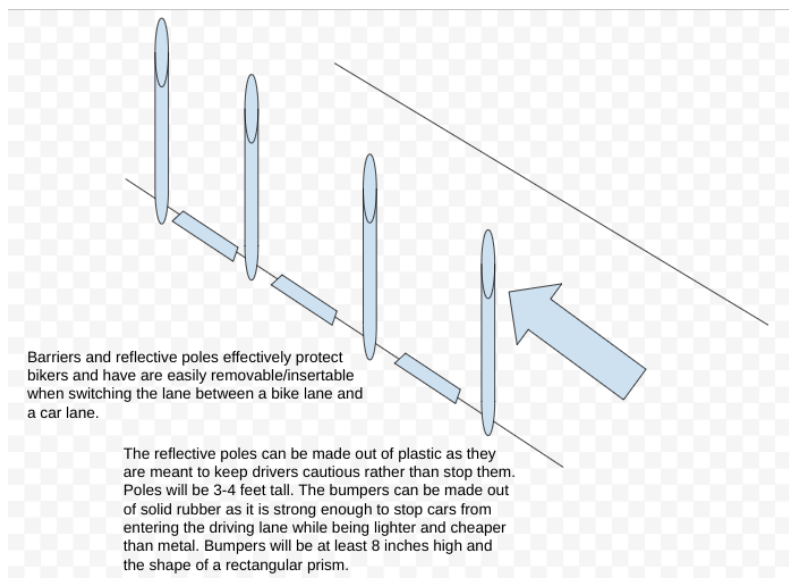
Our team chooses design 1 to solve the bike lane problem because it has safety bollards and lane lines, has wider bike lanes, significant buffers that are at least 5 ft wide and 3 ft tall, and includes a reflective structure to be seen at night.

Design Blueprint (D4)

Design Blueprint #1

Date: 10/24/25

Sketch or CAD Drawing (orthographic/multiview):



Manufacturing Methods (cut, saw, paste, screw, tie, glue):

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Step 1: Cut out cardboard base that will represent paved road

Step 2: Cut and place bollards and barriers made out of hard plastic or cardboard onto the road that contains holes for them to go in

Step 3: Draw lines using a marker to represent the different lanes

Supply List and Costs:

Item	Cost each	# needed	Total
Cardboard	\$0.32	3	\$0.96
Hard Plastic	\$4.49	1	\$4.49
Tape	\$3.50	1	\$3.50
Total Cost			\$8.95

Plan of Action (D3)

- <https://docs.google.com/document/d/1W35fYbW1Z-ybq7TGrycm1kNHqT8Tyve9Qr-VDsftFA/edit?usp=sharing>



Wait until you create and test your first prototype to fill out the next section.

Iterations for the Solution (How did you change your prototype after building and testing it?):

Date of Iteration:

Description of Iteration:

Images if applicable:

Date of Iteration:



Name: _____

Date: _____

Class: _____

Description of Iteration:

Images if applicable:

Date of Iteration:

Description of Iteration:

Images if applicable:

Rubric

Element D: Design Concept Generation, Analysis and Selection						
	5	4	3	2	1	0
1. Process for Generating & Comparing Possible Solutions. The Process for Generating Possible Solutions _____.	was <i>comprehensive,</i> <i>iterative,</i> <i>and</i> <i>consistently</i> defensible (viable design highly likely)	was <i>thorough,</i> <i>iterative,</i> <i>and</i> generally defensible (viable design likely)	was <i>adequate,</i> <i>generally</i> iterative and defensible (<i>viable design</i> <i>is possible</i>)	<i>was partial or</i> <i>overly general</i> <i>and only</i> <i>somewhat</i> iterative and/or defensible (raising issues with the design viability)	<i>was</i> <i>incomplete</i> <i>or minimally</i> <i>iterative</i> <i>and/or</i> <i>defensible</i> (<i>raising</i> <i>issues with</i> <i>the design</i> <i>viability</i>)	<i>was not evident</i> <i>that there was</i> <i>an attempt</i>



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2.Design Solution Choice. The Design Solution chosen _____	<i>was well-justified and demonstrated attention to all design requirements</i>	<i>was justified and demonstrated attention to most if not all design requirements</i>	<i>was explained with reference to at least some design requirements</i>	<i>was not sufficiently explained with reference to design requirements</i>	<i>lacked support related to design requirements</i>	<i>was not attempted or evidence not provided</i>
3. Plan of Action. The plan of action _____	<i>has considerable merit and would easily support repetition and testing for effectiveness by others</i>	<i>would support repetition and testing for effectiveness by others</i>	<i>might not clearly or fully support repetition and testing for effectiveness by others</i>	<i>has insufficient detail to allow for testing for replication of results</i>	<i>is lacking to the point that the design solution cannot be tested</i>	<i>cannot be executed or does not exist</i>